

Roles of extracellular vesicles from mesenchymal stem cells in regeneration

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Abstract

Mesenchymal stem cells (MSCs) are highly valued in regenerative medicine due to their ability to self-renew and differentiate into various cell types. Their therapeutic benefits are primarily due to their paracrine effects, in particular through extracellular vesicles (EVs), which are related to intercellular communication. Recent advances in EV production and extraction technologies highlight the potential of MSC-derived EVs (MSC-EVs) in tissue engineering and regenerative medicine. MSC-EVs offer several advantages over traditional cell therapies, including reduced toxicity and immunogenicity compared with whole MSCs. EVs carrying functional molecules such as growth factors, cytokines, and miRNAs play beneficial roles in tissue repair, fibrosis treatment, and scar prevention by promoting angiogenesis, skin cell migration, proliferation, extracellular matrix remodeling, and reducing inflammation. Despite the potential of MSC-EVs, there are several limitations to their use, including variability in quality, the need for standardized methods, low yield, and concerns about the composition of EVs and the potential risks. Overall, MSC-EVs are a promising alternative to cell-based therapies, and ongoing studies aim to understand their actions and optimize their use for better clinical outcomes in wound healing and skin regeneration.